

## Solution

### FORCE-03

#### Class 09 - Science

1. Yes, an object may travel with a non-zero velocity even when the net external force on it is zero. A raindrop falls down with a constant velocity. The weight of the drop is balanced by the up thrust and the velocity of air. The net force on the drop is zero. This can happen under the following conditions:

- The object should already be traveling with a uniform speed in a straight-line path.
- There should be no change in the magnitude of speed.
- There should be no change in the direction of motion.
- The friction between the object and the ground must be zero.

2. Let, Mass of the first block =  $m_1$

Acceleration of the first block =  $a_1$

Mass of the second block =  $m_2$

Acceleration of the second block =  $a_2$

Also,  $a_2 = 5 a_1$

If the force acting on each block be  $F$ .

Then,  $F = m_1 a_1$  and  $F = m_2 a_2 = m_2 \times 5 a_1$

or  $m_1 a_1 = m_2 a_2 = m_2 \times 5 a_1$

or  $m_1 = 5 m_2$

This gives,  $\frac{m_2}{m_1} = \frac{1}{5}$  Thus, the mass of the second block is one-fifth that of the first block.

3. According to the second law of motion,  $F = ma$ ------(i)

If no external force is acting on the object at rest, we can write,

$F = 0$ ------(ii)

From (i) and (ii), we get

$$m \times a = 0$$

Since,  $m$  can't be zero,  $a=0$ .

According to Newton's second law of motion

$F =$  rate of change of momentum

$$F = \frac{mv - mu}{t}$$

If  $F=0$ , then

$$\frac{mv - mu}{t} = 0$$

or,  $v - u = 0$  ( ' $m$ ' can't be 0)

so,  $v = u$  (for whatever time  $t$  is taken.)

This means that if no external force is applied on the moving object, then its initial and final velocities are equal. It means that the body will continue to be in the state of uniform motion along a straight line if no external force acts on it.

4. Initial velocity of car =  $45 \text{ km h}^{-1}$

$$= 45 \times \frac{5}{18} = \frac{225}{18} \text{ ms}^{-1}$$

Final velocity of the car ( $v$ ) = 0

Time ( $t$ ) = 5 s

Mass of the car ( $m$ ) = 1000 kg

Force exerted by the tree on the car

$$F = \frac{m(v-u)}{t} = \frac{1000 \times \left(0 - \frac{225}{18}\right)}{5}$$
$$= \frac{-1000 \times 225}{90} = -2500 \text{ N}$$

Hence force exerted by the car on tree

$$= -(F) = -(-2500 \text{ N}) = 2500 \text{ N}$$

5. Initial mass of rifle = 4 kg

Let initial velocity of recoil of rifle =  $v$

Q Initial momentum of the rifle =  $4 \times v$

Initial mass of bullet = 50 g

$$= \frac{50}{1000} \text{ kg} = 0.05 \text{ Kg}$$

Initial velocity of bullet =  $35 \text{ ms}^{-1}$

Initial momentum of bullet =  $0.05 \times 35$

Using the law of conservation of momentum;

Momentum of the rifle = Momentum of the bullet

$$4 \times v = 0.05 \times 35$$

$$v = \frac{0.05 \times 35}{4}$$

$$v = 0.4375 \text{ ms}^{-1}$$

6. The working of the rotation of sprinkler is based on third law of motion. As the water comes out of the nozzle of the sprinkler, an equal and opposite reaction force comes into play. So, the sprinkler starts rotating.

7.  $m = 500 \text{ g} = 0.5 \text{ kg}$ ,  $t = 10 \text{ s}$ ,  $F = 10^{-2} \text{ N} = 0.01 \text{ N}$  and  $u = 0$

$$\text{From } F = m a \text{ we have } a = \frac{F}{m} = \frac{0.01}{0.5} = 0.02 \text{ ms}^{-2}$$

$$S = ut + \frac{1}{2} at^2$$

$$\text{We have } S = 0 + \frac{1}{2} \times 0.02 \times (10)^2 = 1 \text{ m}$$

8. When a cart is moving on the road, it has to encounter friction. To maintain a constant speed, some force need to be applied continuously to overcome the friction. Hence, the horse needs to continuously apply a force in order move the cart with a constant speed.

9. Given:  $m = 80 \text{ kg}$ ,  $u = 5 \text{ ms}^{-1}$   $v = 25 \text{ ms}^{-1}$  and  $t = 2 \text{ s}$

$$\text{Now acceleration } a = \frac{\text{change in velocity}}{\text{time taken}} = \frac{v-u}{t} = \frac{25-5}{2} = 10 \text{ ms}^{-2}$$

Force = mass  $\times$  acceleration or  $F = ma$

$$\text{Therefore, } F = 80 \times 10 = 800 \text{ N}$$

10. It is based on the law of conservation of momentum. When water comes out of the hose, with certain momentum in the forward direction, the hose, in order to conserve momentum moves backward. This makes it difficult for the fireman to hold the hose.

11. We know,  $F = ma = 5 \text{ N}$  or  $5 \text{ kg ms}^{-2}$

$$\Rightarrow m_1 = \frac{F}{a_1} = \frac{5}{8} \text{ kg}$$

$$\Rightarrow m_2 = \frac{F}{a_2} = \frac{5}{24} \text{ kg}$$

If the mass are tied together, then

$$\therefore M = \left( \frac{5}{8} + \frac{5}{24} \right) \text{ kg} = \left( \frac{5}{6} \right) \text{ kg}$$

Acceleration produced in M,

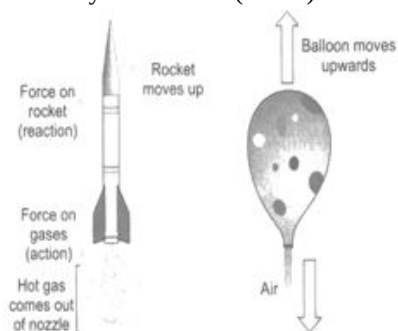
$$a = \frac{F}{M} = \frac{5}{\frac{5}{6}} = 6 \text{ ms}^{-2}$$

Therefore, if both the masses are tied together, the force would result in an acceleration of  $6 \text{ ms}^{-2}$ .

12. a. Jet airplanes and rockets work on the principle of Newton's third law of motion.

In this case, the hot gases come out of a nozzle (a fine opening) with a great force, i.e., action and the rocket moves with high speed upwards as a reaction.

b. If we fill a balloon with air and hold it with its mouth downwards, then when we release the balloon, the air rushes out vertically downwards (action). \*The balloon moves vertically upwards (reaction).



c. When a gun is fired, the bullet goes out (action) due to the force applied on it. According to Newton's third law of motion, the gun recoils backward due to the reaction acting on it in the opposite direction.



13. We can explain the above observation on the basis of law of conservation of momentum. When a system of the hose and water is not ejecting any water its momentum is zero. When the water issues out from the hose with a high velocity, it has momentum in the forward direction. Therefore, in order to conserve momentum the hose tends to move in the backward direction and hence is difficult to hold.

14. Mass of ball (m) = 200 g = 0.2 kg

Initial velocity of ball ( $u_1$ ) =  $10 \text{ ms}^{-1}$

Final velocity of ball ( $u_2$ ) =  $-5 \text{ ms}^{-1}$

(Negative sign denotes that ball is moving in opposite direction)

Initial momentum of ball =  $mu_1$

=  $0.2 \times 10 = 2 \text{ Ns}$

Final momentum of ball =  $mu_2$

=  $0.2 \times (-5) = -1 \text{ Ns}$

Change in Momentum = Final Momentum - Initial Momentum

=  $-1 - 2 = -3 \text{ Ns}$

Negative sign denotes that change in momentum is in the direction opposite to the direction of initial momentum of the ball.

15. We are given,

Acceleration,  $a = 8 \text{ ms}^{-2}$

Initial velocity,  $u = 0$

Time interval,  $t = 1 \text{ s}$

Distance travelled,  $S = ?$

Using the equation of motion,  $S = ut + \frac{1}{2}at^2$ , one gets

$S = 0 \times 1 + \frac{1}{2} \times 8 \times 1^2$

The object travels a distance of 4 m.

16. In a car accident, a fast running car stops suddenly. Due to this the car's large momentum is reduced to zero in a very short time. The stretchable seat belts tightened by the passengers of the car increases the time taken by the passengers to fall forward. Due to long time, the rate change of momentum of passengers is reduced and hence less stopping force acts on them. So, the passengers may either not get injured at all or may get less injuries. It is obvious that seat belts reduce the passengers' momentum more gently and hence prevent injuries.

17. It is given that  $u = 3 \text{ ms}^{-1}$  and  $v = 7 \text{ ms}^{-1}$ ,  $t = 2 \text{ s}$  and  $m = 5 \text{ kg}$

Now according to Newton's 2nd Law of motion

$$F = \frac{m(v-u)}{t}$$

Substitution of values in this relation gives

$$F = 5 \text{ kg} \frac{(7 \text{ ms}^{-1} - 3 \text{ ms}^{-1})}{2 \text{ s}} = 10 \text{ N}$$

Now, if this force is applied for a duration of 5 s ( $t = 5 \text{ s}$ ), then the final velocity can be calculated as

$$v = u + \frac{Ft}{m}$$

On substituting the values of  $u$ ,  $F$ ,  $m$ , and  $t$ , we get the final velocity,  $v = 13 \text{ ms}^{-1}$ .

18. Mass of bullet (m) = 10 g = 0.01 kg

Initial velocity of the bullet ( $u$ ) = 0

Final velocity of the bullet ( $v$ ) =  $300 \text{ ms}^{-1}$

Time ( $t$ ) = 0.003 s

Now acceleration of the bullet  $a = \frac{v-u}{t} = \frac{300-0}{0.003} = 100,000 \text{ ms}^{-2}$

Hence force on the bullet  $F = ma$

$F = 0.01 \times 100,000 = 1000 \text{ N}$

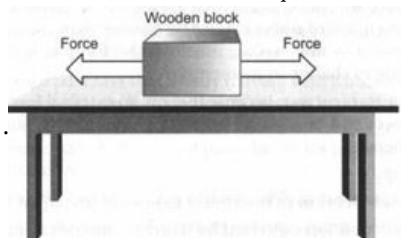
19. Luggage on the top of the bus is a loose fixture and not a compact part of the bus. Thus, when a speeding bus brakes suddenly, the luggage continues moving forward because of inertia of motion and is likely to fall off the bus. Conversely, if a stationary bus

accelerates suddenly, the luggage continues in the same state because of inertia of rest and hence is left behind relative to bus such that it falls backward. To avoid the falling of luggage, it is tied with a rope.

20. Quantum of inertia depends on mass of the object. For same volume, the mass of steel would be highest; compared to aluminium or wood. Hence, the solid of steel would have the highest inertia.
21. The bus and the persons in it are in the state of motion and at rest with respect to each other before the application of the brakes. When the brakes are suddenly applied, the bus comes to the state of rest, but the passengers tend to continue in the state of motion because of inertia of motion. Therefore, they fall in forward direction. Conversely, when the bus is in the state of rest, the passengers in it have inertia of rest. When the bus accelerates suddenly, the passengers tend to continue in their state of rest and hence are left behind, relative to the position of bus. Therefore, they fall in the backward direction.
22. Mass of the vehicle ( $m$ ) = 1500 kg  
Negative acceleration ( $a$ ) =  $-1.7 \text{ ms}^{-2}$   
 $\therefore$  Force of friction between the road and vehicle  
 $F = ma$   
 $F = 1500 \times (-1.7)$   
 $= -2550 \text{ N}$   
Negative sign means force is acting in the direction opposite to the direction of motion of the vehicle.
23. It is on account of inertia of motion. The athlete runs a considerable distance so as to build up momentum, which is helpful in throwing the javelin at a longer distance. However, sometimes the large momentum of athlete prevents him from stopping before the marked line therefore the throw is declared foul.
24. When we walk on the ground or road, our foot pushes the ground in backward direction (i.e. action). According to Newton's third law of motion, for every action, there is equal and opposite reaction. Therefore, the ground pushes our foot in forward direction (i.e. reaction). Thus, the forward reaction exerted by the ground on our foot makes us walk in forward direction.
25. Yes, the balls will start rolling in the direction in which the train was moving. Due to the application of the brakes, the train comes to rest but due to inertia the balls try to remain in motion, therefore, they begin to roll. Since the masses of the balls are not the same, therefore, the inertial force are not same on both the balls. Thus, the balls will move with different speeds.
26. If the resultant of various forces acting on a body is zero, the forces are said to be 'balanced forces'. These forces do not change the speed but usually change the shape of an object. Examples:

- a. Consider a wooden block lying on a table, the strings tied to its two opposite faces, as shown in the figure.

If we pull at point P, it begins to move towards left. If we pull at point Q, it begins to move towards the right. But if we pull from both the sides with equal force, the block does not move. The two forces have now balanced each other



- b. In a tug-of-war, the two teams pull the rope with equal effort; the rope is not moved in any direction. This is clearly because the forces exerted by the two teams are equal and opposite and thus get balanced.
27. An athlete runs a certain distance before taking a leap so that the inertia of motion of his body at the time of leaping may help him in his muscular efforts.
28. The player does so as to decrease the rate of change of momentum by increasing the time. In doing so, the entire momentum of the ball is reduced to zero in a long time interval. In other words, the rate of change of momentum is small. As a result, according to Newton's second law of motion, the player has to apply a small force on the ball. In reaction, the ball also applies less force and the palms of the player are not injured. If the ball is stopped suddenly, then the entire momentum of the ball will be reduced to zero in a very short time which will cause a larger rate of change of momentum resulting in greater action and reaction forces. Thus the palms of the player will be hurt.
29. When a small mass such as a bullet is fired from a gun, it moves out of the gun with a very high velocity. Thus, the momentum produced by a bullet is very high ( $p = mv$ ). This high momentum of the bullet can kill a person.
30. i. Inertia depends on the mass of a body.  
ii. Momentum depends on mass and velocity of the body.  
iii. Force depends on the mass and acceleration of the body.

31. When a bullet fired against the glass pane, the bullet strikes the glass pane with very high velocity. The part of the glass pane which comes in contact with the bullet immediately shares the large velocity of the bullet and makes a hole in the glass pane, while the remaining part of the glass remains at rest due to inertia of rest and is therefore not cracked.  
But, on the other hand, when a slow-moving stone strikes the same glass pane, the various parts of the glass pane gets enough time to share the velocity of the stone. Therefore, the glass is smashed.
32. Mass of stone (m) = 1 kg  
Initial velocity of stone (u) = 20 ms<sup>-1</sup>  
Final velocity of stone (v) = 0  
Distance covered by the stone (s) = 50 m  
Acceleration of stone (a) = ?  
Force acting on the stone due to friction (F) = ?  
We know;  
 $v^2 - u^2 = 2as$   
 $(0)^2 - (20)^2 = 2a \times 50$   
 $-400 = 100a$   
 $a = \frac{-400}{100} = -4$   
 $a = -4 \text{ ms}^{-2}$   
Force of friction (F) = ma  
 $= 1 \times (-4)$   
 $= -4 \text{ N}$   
Negative sign signifies that force of friction is acting in the direction opposite to the direction of motion of the stone.
33. When the branch of a tree is shaken vigorously, it comes in motion while the leaves tend to remain at rest due to inertia of rest. Therefore, some of the leaves get detached from the branch of the tree.
34. When the branch is suddenly set in motion, the leaves attached to it tend to continue in their state of rest because of inertia of rest. It generates a lot of strain at the junction of leaves and the branch. Because of this strain some leaves get detached from the branch.
35. We have given that,  
The initial velocity of the motorcar u = 108 km/h  
 $= \frac{108 \times 1000 \text{ m}}{(60 \times 60 \text{ s})}$   
 $= 30 \text{ ms}^{-1}$   
and the final velocity of the motorcar v = 0 ms<sup>-1</sup>  
The total mass of the motorcar along with its passengers = 1000 kg and the time taken to stop the motorcar, t = 4 s.  
We have the magnitude of the force (F) applied by the brakes as  
 $\frac{m(v-u)}{t}$   
On putting the values in the above equation, we get  
 $F = 1000 \text{ kg} \times (0 - 30) \text{ ms}^{-1} = \frac{1000 \text{ kg} \times (0 - 30) \text{ ms}^{-1}}{4 \text{ s}}$   
 $= -7500 \text{ kg ms}^{-2}$  or -7500 N  
The negative sign tells us that the force exerted by the brakes is opposite to the direction of motion of the motorcar.
36. In the given example the velocity of football changes four times. It is described below:  
(i) when the football player is supplying the force when he kicks the football to another player.  
(ii) when the other player kicks football towards the goal.  
(iii) When the goalkeeper of other team stops the football.  
(iv) When the goalkeeper kicks the football towards player of his team.  
Agents supplying the force:  
Case (i)-First player.  
Case (ii)-Second player  
Case (iii)-Goal keeper  
Case (iv)- Goal keeper.
37. Mass of the body (m) = 5 kg  
Force acting on the body (F) = 200 N

$$\therefore \text{Acceleration acting on the body} = a = \frac{F}{M} = \frac{200}{5} = 40 \text{ ms}^{-2}$$

38. Here,  $m = 0.4 \text{ kg}$ ,  $u = 10 \text{ m/s}$

Initial momentum of the ball =  $mu = 0.4 \times 10 = 4 \text{ kg m/s}$

At the highest point, velocity of ball is zero,

Therefore, the momentum of the ball at the highest point of flight =  $0 \times 4 = 0$ .

39. When we walk on the ground the force of friction between the sole of our shoe and the ground acts opposite to the direction of our motion. It is this force of friction which gradually wears out the sole of our shoe.

40. According to the question,

Initial velocity of truck ( $u$ ) = 0

Distance,  $S = 400 \text{ m}$  and time,  $t = 20 \text{ s}$

Mass of truck,  $m = 7 \text{ metric tones} = 7000 \text{ kg}$

From the relation,

$$S = ut + \frac{1}{2}at^2$$

$$\Rightarrow 400 = 0 + 200a$$

$$\Rightarrow 400 = 200a$$

$$\Rightarrow a = 400/200 = 2 \text{ ms}^{-2}$$

Thus, the acceleration of the truck is  $2 \text{ ms}^{-2}$

Therefore,  $F = m \times a = 7000 \times 2 = 14000 \text{ N}$

Thus, the force acting on the truck is  $14000 \text{ N}$

41. Here, the mass of the hammer =  $500 \text{ g} = \frac{500}{1000} \text{ kg} = 0.5 \text{ kg}$ ,

Initial velocity,  $u = 50 \text{ ms}^{-1}$

Final velocity,  $v = 0$

Time taken,  $t = 0.01 \text{ s}$

From Newton's second law of motion, we have

$$F = ma = \frac{m(v-u)}{t} = \frac{0.5 \text{ kg}(0 \text{ ms}^{-1} - 50 \text{ ms}^{-1})}{0.01 \text{ s}}$$

$$= \frac{-25 \text{ kg ms}^{-1}}{0.01 \text{ s}}$$

$$= -2500 \text{ kg ms}^{-2} = -2500 \text{ N}$$

So, the force of the nail on the hammer is  $2500 \text{ N}$ . Negative sign indicates that the force is acting opposite to the motion.

42. Mass of bullet ( $m$ ) =  $10 \text{ g} = 0.01 \text{ kg}$

Initial velocity of bullet ( $u$ ) =  $150 \text{ ms}^{-1}$

Final velocity of bullet ( $V$ ) = 0

Time ( $t$ ) =  $0.03 \text{ sec}$

Acceleration on bullet ( $a$ ) = ?

Force acting on wooden block ( $F$ ) = ?

Distance penetrated by bullet ( $s$ ) = ?

We know:

$$v = u + at$$

$$0 = 150 + a \times 0.3$$

$$\Rightarrow -a \times 0.03 = 150$$

$$\Rightarrow a = \frac{-150}{0.03} = -5000 \text{ ms}^{-2}$$

We know;  $S = ut + \frac{1}{2}at^2$

$$= 150 \times 0.03 + \frac{1}{2} \times (-5000) \times (0.03)^2$$

$$= 4.5 - 2.25 = 2.25 \text{ m}$$

We know,  $F = ma$

Force acting on bullet ( $F$ )

$$= 0.01 \times (-5000)$$

$$= -50 \text{ N}$$

Negative sign denotes that wooden block exerts force in the direction, opposite to the direction of motion of the bullet.

43. Initially, the carpet and dust-particles are at rest. When the carpet is beaten, it is suddenly set into motion. The dust particles tend to remain at rest due to the inertia of rest, therefore, dust comes out of it.

44. We have,

$$\text{Mass of bullet, } m_1 = 50g = \frac{50}{1000}kg = 0.05kg$$

$$\text{Mass of rifle, } m_2 = 4kg$$

Before firing,

$$\text{velocity of bullet, } u_1 = 0$$

$$\text{velocity of rifle, } u_2 = 0$$

$$\text{After firing, velocity of bullet } v_1 = 35ms^{-1}$$

Let the recoil velocity of the rifle be  $v_2$

Before firing, both the gun and the bullet are at rest. So, the total momentum of the system before firing = 0

$$\text{i.e., } m_1u_1 + m_2u_2 = 0.$$

Total momentum of the system after firing

$$= m_1v_1 + m_2v_2$$

$$= 0.05kg \times 35ms^{-1} + 4kg \times v_2$$

$$= 1.75kgms^{-1} + 4kg \times v_2$$

As there is no external force, so according to the law of conservation of momentum,

Total momentum of the system before firing = Total momentum of the system after firing

$$\Rightarrow 1.75kgms^{-1} + 4kg \times v_2 = 0$$

$$\Rightarrow v_2 = -\frac{1.75kgms^{-1}}{4kg}$$

$$\Rightarrow v_2 = -0.4375m/s = -0.44m/s$$

Negative sign indicates that the direction in which the rifle would recoil is opposite to that of bullet.

45. Given mass = 1,200 kg, acceleration produced =  $0.2ms^{-2}$

Force exerted by two persons simply overcome the force of friction between the ground and the motor car. The force of third person imparts acceleration to the motor car; so force exerted by third person,  $F = ma = 1,200 \times 0.2 = 240N$

$\therefore$  Force exerted by each person = 240 N

46. Acceleration can be given as follow:

$$F = ma \text{ or } a = \frac{f}{m} \text{ or } a_1 = \frac{F}{M}$$

When mass is doubled and force is halved;

$$a_2 = \frac{F}{4M} \text{ or } \frac{a_2}{a_1} = \frac{F}{4M} \div \frac{F}{M} = \frac{1}{4}$$

So, acceleration becomes one-fourth.

47. We are given,

$$\text{Initial velocity } u = 0ms^{-1}$$

$$\text{Distance travelled, } S = 400m$$

$$\text{Time interval, } t = 20s$$

$$\text{Mass of the truck, } m = 7\text{ m ton} = 7000kg$$

a) To find the acceleration of the truck

Using the equation of motion,  $S = ut + \frac{1}{2}at^2$  we have

$$400 = 0 + \frac{1}{2}a(20)^2 \Rightarrow a = \frac{2 \times 400}{20 \times 20} = 2ms^{-2}$$

b) To find the force acting on the truck

Using Newton's second law of motion,

$$F = m \times a = 7000 \times 2ms^{-2} = 14000N$$

48.  $u = 80m/s$ ,  $v = 0m/s$  and  $t = 8s$

Acceleration can be calculated as follows:

$$v = u + at$$

$$\text{Or } a = \frac{v-u}{t} = -\frac{80}{8}ms^{-2} = -10ms^{-2}$$

Frictional force can be calculated as follows:

$$F = m \times a = \frac{50}{1000}kg \times 10ms^{-2} = 0.5N$$

49. Let  $m$  be the mass of the body

$$\text{Given : } F = 40N, a = 5ms^{-2}$$

From the relation :  $F = ma$  we have  $40 = m \times 5$

$$\text{or } m = \frac{40}{5} = 8kg$$

50. Here,  $m_1 = 2 \text{ kg}$ ,  $a_1 = 5 \text{ ms}^{-2}$ ,  $m_2 = 4 \text{ kg}$ ,  $a_2 = 2 \text{ ms}^{-2}$

$$F_1 = m_1 a_1 = 2 \times 5 = 10 \text{ N}$$

$$F_2 = m_2 a_2 = 4 \times 2 = 8 \text{ N}$$

$$F_1 > F_2$$

Thus, accelerating a  $2 \text{ kg}$  mass at  $5 \text{ ms}^{-2}$  acceleration would require a greater force than accelerating a mass of  $4 \text{ kg}$  at  $2 \text{ ms}^{-2}$ .

51. The initial velocity of the ball is  $20 \text{ cm s}^{-1}$ .

Due to the frictional force exerted by the table, the velocity of the ball decreases down to zero in  $10 \text{ s}$ .

Thus,  $u = 20 \text{ cm s}^{-1}$ ;  $v = 0 \text{ cm s}^{-1}$  and  $t = 10 \text{ s}$ . Since the velocity-time graph is a straight line, it is clear that the ball moves with a constant acceleration. The acceleration 'a' is given by

$$\begin{aligned} a &= \frac{v-u}{t} \\ &= \frac{(0 \text{ cm s}^{-1} - 20 \text{ cm s}^{-1})}{10 \text{ s}} \\ &= -2 \text{ cm s}^{-2} = -0.02 \text{ m s}^{-2} \end{aligned}$$

The force exerted on the ball  $F$  is obtained from Newton's 2nd law of motion

$$\begin{aligned} F &= ma = \left(\frac{20}{1000}\right) \times (-0.02 \text{ m s}^{-2}) \\ &= -0.0004 \text{ N} \end{aligned}$$

The negative sign implies that the frictional force exerted by the table is opposite to the direction of motion of the ball.

52. When the passengers alight from the boat, they push the boat in backward direction. As a result, the boat has a tendency to slip back into water. This difficulty is usually overcome by the boatman by tying the boat to some rigid support.

53. We have  $m_1 = \frac{F}{a_1}$ , and  $m_2 = \frac{F}{a_2}$ .

Here we have given that,  $a = 1 = 10 \text{ ms}^{-2}$ ,  $a_2 = 20 \text{ ms}^{-2}$  and  $F = 5 \text{ N}$

$$\text{Thus } m_1 = \frac{5 \text{ N}}{10} \text{ ms}^{-2} = 0.50 \text{ kg; and } m_2 = \frac{5 \text{ N}}{20} \text{ ms}^{-2} = 0.25 \text{ kg}$$

If the two masses were tied together, the total mass,  $m$  would be  $m = 0.50 \text{ kg} + 0.25 \text{ kg} = 0.75 \text{ kg}$

The acceleration,  $a$  produced in the combined mass by the  $5 \text{ N}$  force would be,  $a = \frac{F}{m} = \frac{5 \text{ N}}{0.75 \text{ kg}} = 6.67 \text{ ms}^{-2}$ .

54.  $m = 5 \text{ g} = 5 \times 10^{-3} \text{ kg}$ ,  $u = 120 \text{ ms}^{-1}$ ,  $v = 0$ ,  $t = 0.01 \text{ s}$

a. From the relation  $v = u + at$

$$\text{We have } 0 = 120 + a \times 0.01$$

$$\text{or } a = -\frac{120}{0.01} = -12000 \text{ ms}^{-2} \text{ (the negative sign here shows retardation)}$$

Distance of penetration in the target

$$S = ut + \frac{1}{2} at^2 \text{ we have}$$

$$S = 120 \times 0.01 + \frac{1}{2} \times (-12000) \times (0.01)^2 = 0.6 \text{ m}$$

b. Average retarding force  $F = ma = (5 \times 10^{-3}) \times (12000) = 60 \text{ N}$

55. The force of nail on the hammer

$$\begin{aligned} F &= \frac{\text{Change in momentum of hammer}}{\text{Time}} \\ F &= \frac{m(v-u)}{t} = \frac{0.50(0-50)}{0.01} \\ &= -2500 \text{ N} \end{aligned}$$

NA negative sign denotes the force of nail on the hammer, acting in the direction, opposite to that of the hammer.

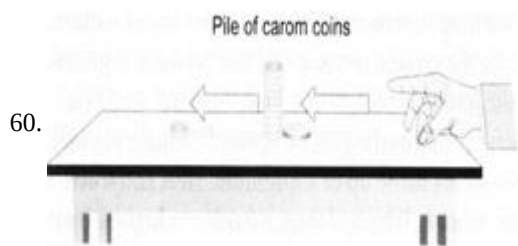
56. The ball slow down and comes to rest due to the opposing forces of air resistance and friction between the ball and the ground.

57. When the standing tanker suddenly picks up speed, the oil in it on account of inertia of rest tends to continue in its state of rest.

Thus, in a way oil is left behind and, hence, exerts very large force on its rear wall. Conversely, when the moving tanker suddenly stops, the oil in it continues moving forward and, hence exerts a very large force on the front wall. These forces can crack the walls of the oil tanker. Therefore to avoid such a mis-hap, some space is left at the top of the tanker for the free movement of oil.

58. Yes, the balls will start rolling in the direction in which the train was moving. Due to the application of the brakes, the train comes to rest but due to inertia the balls try to remain in motion. Therefore, they begin to roll. Since the masses of the balls are not the same, therefore, the inertial forces are not same on both the balls. Thus, the balls will move with different speeds.

59. When the branch is, suddenly set in motion, the leaves attached to it tend to continue in their state of rest, on account of inertia of motion. Thus a lot of strain acts on the junction of the leaves and the branches. Due to this strain the weakly held leaves are left behind and, hence fall off the branch.



- Make a pile of similar carrom coins on a table, as shown in the figure.
- Attempt a sharp horizontal hit at the bottom of the pile using another carrom coin or the striker. If the hit is strong enough, the bottom coin moves out quickly. Once the lowest coin is removed, the inertia of the other coins makes them 'fall' vertically on the table.
- The remaining coins due to the inertia of rest try to remain at rest but fall down due to gravity.

61. Here,

Mass of the automobile,  $m = 1,500 \text{ kg}$

Negative acceleration,  $a = -1.7 \text{ ms}^{-2}$

Force needed to stop the vehicle,  $F = ?$

Using the relation,  $F = ma$

$$= 1500 \text{ kg} \times (-1.7 \text{ ms}^{-2}) = -2,550 \text{ kg ms}^{-2} = -2550 \text{ N}$$

The negative value of the force indicates that it is acting in the direction opposite to the direction of motion. So, the retarding force needed to stop the vehicle is 2550 N.

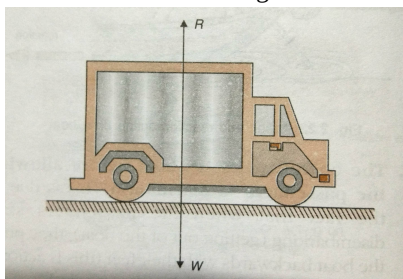
62. a. Net accelerating force is given by
- $$= \text{Force exerted by the engine} - \text{Frictional force}$$
- $$= 40000 - 5000 = 35000 \text{ N}$$

b. Acceleration is given by

$$a = \frac{\text{Accelerating force}}{\text{Mass of train}}$$

$$= \frac{35000 \text{ N}}{10000 + 8000 \text{ kg}} = 1.94 \text{ ms}^{-2}$$

63. The various forces acting on the truck at rest are as follows:



Here, the weight of the truck  $W$  is balanced by the reaction  $R$  of the ground on the truck. But the frictional force due to the ground is much more than the force of push. Therefore, the truck does not move.

64. As we know mass of body is the measure of its inertia, that is more the mass of a body, more is its inertia. So,
- Stone has more inertia than a rubber ball of the same size.
  - Train has more inertia than a bicycle, and
  - A five-rupee coin has more inertia than a one-rupee coin.

65. Here,

Mass,  $m = 1 \text{ kg}$

Initial velocity,  $u = 20 \text{ ms}^{-1}$

Final velocity,  $v = 0 \text{ ms}^{-1}$

Distance travelled,  $S = 50 \text{ m}$

Force of friction,  $F = ?$

From the third equation of motion, we have,  $v^2 = u^2 + 2as$

$$\Rightarrow a = \frac{v^2 - u^2}{2s} = \frac{0 - (20 \text{ ms}^{-1})^2}{2 \times 50 \text{ m}}$$

$$= \frac{-400 \text{ m}^2 \text{ s}^{-2}}{100 \text{ m}} = -4 \text{ ms}^{-2}$$

From Newton's second law of motion, we have

$$F = ma = 1\text{kg} \times (-4\text{ms}^{-2}) = -4\text{kgms}^{-2} = -4\text{ N}$$

Negative sign indicates the force is acting opposite to the direction of motion.

Thus, the force of friction between the stone and ice is -4 N

66. a. a stone of the same size will have more inertia than a rubber ball.  
 b. A train will have more inertia than a bicycle.  
 c. A five rupees coin will have more inertia than a one-rupee coin.

Explanation: Inertia is associated with mass. Inertia is directly proportional to the mass of a body. Objects having more mass have more inertia.

67. i. Initial velocity of the car ( $u$ ) =  $90\text{ kmh}^{-1}$

$$= 25\text{ ms}^{-1}$$

$$\text{Final velocity of the car (v)} = 18\text{ kmh}^{-1} = 5\text{ ms}^{-1}$$

$$\text{Time (t)} = 4\text{s}$$

$$\text{Acceleration} = (a) = ?$$

$$\text{We know: } v = u + at$$

$$5 = 25 + a \times 4$$

$$\therefore -a \times 4 = 20$$

$$\Rightarrow a = \frac{-20}{4} = -5\text{ms}^{-2}$$

- ii. Change in momentum ( $p$ ) =  $m(v - u)$

$$= 1200(5 - 25)$$

$$= 1200 \times (-20)$$

$$= -2400\text{ Ns}$$

$$\text{iii. Magnitude of force (F)} = \frac{m(v-u)}{t} = \frac{24000}{4}$$

$$= 6000\text{ N}$$

68. In a moving bus, a passenger moves with the bus due to inertia of motion. As the driver applies brakes, the bus comes to rest. But, the passenger tries to maintain to inertia of motion. As a result, a forward force is exerted on him and the passenger leans forward.

Similarly, the passenger tends to fall backwards when the bus accelerates from rest because when the bus accelerates, the inertia of rest of the passenger tends to oppose the forward motion of the bus. Hence, the passenger tends to fall backwards when the bus accelerates forward.

69. i.  $m = 10\text{g} = \frac{10}{1000}\text{kg}$ ,  $u = 10^3\text{ m/s}$ ,  $v = 0$ ,  $s = \frac{5}{100}\text{m}$

$$v^2 - u^2 = 2as$$

$$\Rightarrow 0 - (10^3)^2 = 2.a. \frac{5}{100}$$

$$\Rightarrow a = \frac{-1000 \times 1000}{2 \times 5} \times 100$$

$$= -10^7\text{ ms}^{-2}$$

$$\Rightarrow F = m.a = -10^5\text{ N}$$

- ii.  $v = u + at$

$$\Rightarrow 0 = (10^3) - 10^7 t$$

$$\Rightarrow 10^7 t = 10^3$$

$$\Rightarrow t = \frac{10^3}{10^7} = 10^{-4}\text{ s}$$

70. Initial velocity ( $u$ ) = 0, Final velocity ( $v$ ) = ?

$$\text{Height (S)} = 0.8\text{ m, Acceleration (g)} = 10\text{ ms}^{-2}$$

$$\text{Using, } v^2 - u^2 = 2gS \text{ we have}$$

$$v^2 - (0)^2 = 2 \times 10 \times 0.8$$

$$v^2 = 16 \text{ or } v = 4\text{ ms}^{-1}$$

$$\text{Therefore, Momentum of dumbbell} = m \times v$$

$$= 10\text{ kg} \times 4\text{ ms}^{-1} = 40\text{ N.}$$

71. According to Newton's third law of motion, 'To every action, there is an equal and opposite reaction; action and reaction forces act on different bodies.'

When a person walks on the ground, then he pushes the ground backwards (action) with a force  $F$  and in reaction the ground also

pushes his foot(reaction) in the forward direction and hence the man walks forward.

According to Newton's third law of motion:- Every action has an equal and opposite reaction.

72. Here, Initial velocity of sphere,  $u = 0$

$h = 10 \text{ m}$

Acceleration of sphere,  $a = 9.8 \text{ ms}^{-2}$

Final velocity of sphere when it just reaches the ground can be calculated using

$$v^2 - u^2 = 2gh$$

$$\implies v^2 - 0 = 2 \times 9.8 \text{ ms}^{-2} \times 10 \text{ m} = 196 \text{ m}^2 \text{ s}^{-2}$$

$$\implies v = \sqrt{196 \text{ m}^2 \text{ s}^{-2}} = 14 \text{ ms}^{-1}$$

Momentum of the sphere just before it touches the ground =  $mv$

$$= 1 \text{ kg} \times 14 \text{ ms}^{-1} = 14 \text{ kgms}^{-1}$$

On reaching the ground, the iron sphere comes to rest, so its final momentum is equal to zero according to the law of conservation of momentum.

Momentum transferred to the ground = Momentum of the sphere just before it comes to rest =  $14 \text{ kgms}^{-1}$

73. In the course of walking on the hard ground, as we push the earth backward with our feet, the earth applies a reaction and pushes us in the forward direction. However, when the reaction is missing we tend to fall. This often happens when we step on a banana peel. The peel slips under our feet and hence does not apply any reaction. Thus, we lose our balance and fall.